

Optimization with CasADi

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Optimization

$$\min_w J(w)$$

Example #1 – Linear Programming

Farmer wants to grow **apples** and **bananas**.

	Area needed [m ² /ton]	Fertilizer needed [kg/ton]	Profit [USD/ton]
A	4000	60	700
B	3000	80	600

Farmer has a field of 100 000 m² and can legally use no more than 2000 kg fertilizer.

What to plant to maximize profit?

Example #1 – Formulation of OCP

m_a , m_b – mass of apple, and banana

Control variables,

$$w = [m_a, m_b]$$

the profit,

$$P(w) = 700m_a + 600m_b$$

the area used,

$$A(w) = 4000m_a + 3000m_b$$

fertilizer used,

$$F(w) = 60m_a + 80m_b$$

$$\max_w P(w) \rightarrow \min_w [-P(w)]$$

$$\text{s.t. } 0 \leq A(w) \leq 100000$$

$$0 \leq F(w) \leq 2000$$

Optional constraints:

$$0 \leq m_a, m_b$$

Example #1 – Solution in LibreOffice Calc

You go to menu Tools/Solver

	A	B	C	D	E	F	G	H
1		mass to plant (ton)						
2	Apple	14.2857142857143						
3	Banana	14.2857142857143						
4								
5	profit	18571.4285714286						
6								
7	s.t.							
8	area needed	100000	<=	100000				
9	fertilizer needed	2000	<=	2000				
10								
11								
12								
13								
14								
15								
16								

Solver

Target cell: **\$B\$5**

Optimize result to: ☒ Maximum
☐ Minimum
☐ Value of

By changing cells: **\$B\$2:\$B\$3**

Limiting Conditions

Cell reference	Operator	Value
\$B\$8	≤	\$D\$8
\$B\$9	≤	\$D\$9
	≤	
	≤	

Help Reset All Options... Close Solve